

# Use Python to build *AI Chatbots* *and AI Agents*



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# LangChain

We will use LangChain to build AI Chatbots and Applications. LangChain is a framework for developing AI applications in Python. It simplifies every stage of the AI application lifecycle

# Let's see how Langchain *works*



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# Build AI Chatbot

The following codes will create an AI Chatbot using Google Gemini

```
from langchain_google_genai import ChatGoogleGenerativeAI
chat = ChatGoogleGenerativeAI(
    model="gemini-pro",
    google_api_key=google_api_key,
    temprature=0.1
)
```

# Build AI Agents

Here is how to create an AI agent that could perform analysis in Python

```
from langchain.agents.agent_types import AgentType
from langchain_experimental.agents.agent_toolkits import create_pandas_dataframe_agent
return create_pandas_dataframe_agent(
    chat,
    data,
    verbose=True,
    max_iterations = 5,
    handle_parsing_errors=True
)
```

# Did you notice?

We used the AI Chatbot model for AI agent. The reason is that the core of the AI Agent uses the same LLM model as chatbot.

Let's see both  
*in action*



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# AI Chatbot

Here is how to ask question to AI

```
to_markdown(  
    chat.invoke("what is AI?")  
    .content  
)
```

chat.invoke is used to ask the question to AI and to\_markdown is a helper function to prettify the output

# AI Chatbot

Here is the output of our question.

Artificial intelligence (AI) refers to the simulation of human intelligence processes by machines, particularly computer systems. It involves the development of computer systems that can perform tasks normally requiring human intelligence, such as:

- **Learning:** Acquiring knowledge and insights from data and experiences.
- **Reasoning:** Applying logic and knowledge to solve problems and make decisions.
- **Planning:** Developing strategies and actions to achieve specific goals.
- **Natural language processing:** Understanding and generating human language.
- **Computer vision:** Interpreting and making sense of images and videos.
- **Robotics:** Designing and controlling physical robots to perform various tasks.

AI is often classified into different types based on its capabilities and the techniques used:

- **Machine learning:** Algorithms that enable computers to learn from data without explicit programming.
- **Deep learning:** A type of machine learning that uses artificial neural networks to process data.
- **Natural language processing (NLP):** Techniques for understanding and generating human language.
- **Computer vision:** Algorithms for interpreting and making sense of images and videos.
- **Robotics:** The design, construction, and control of physical robots.

AI has numerous applications in various industries and domains, including:

- **Healthcare:** Diagnosis, treatment, and drug discovery.
- **Finance:** Fraud detection, risk assessment, and trading.
- **Transportation:** Self-driving cars, traffic management, and logistics.
- **Manufacturing:** Automation, quality control, and predictive maintenance.
- **Customer service:** Chatbots, recommendation systems, and personalization.
- **Education:** Personalized learning, assessment, and virtual assistants.
- **Entertainment:** Video games, virtual reality, and music generation.

AI is rapidly evolving, with advancements in algorithms, computing power, and data availability. It has the potential to transform many aspects of society, from healthcare to finance to transportation. However, it also raises important ethical, societal, and economic considerations that need to be addressed as AI continues to develop.

# AI Agent

Here is the data I have passed to the AI agent to perform analysis on.

|   | name                                              | revenue |
|---|---------------------------------------------------|---------|
| 0 | The North Face Apex Bionic Soft Shell Jacket -... | 4515.0  |
| 1 | Darla                                             | 3996.0  |
| 2 | Jordan Low Quarter Sock Style # 427411            | 3612.0  |
| 3 | Canada Goose Men's The Chateau Jacket             | 3260.0  |
| 4 | Canada Goose Men's Citadel Parka                  | 3180.0  |

# AI Agent

Here is how to ask analysis questions  
an agent

```
to markdown(  
    agent.invoke("""WHich product is better for the  
    business in terms of revenue?""")  
    ["output"]  
)
```

agent.invoke is used here as well similar to AI  
Chatbot

# AI Agent

Here is the output of our question:

> Finished chain.

The North Face Apex Bionic Soft Shell Jacket - Men's

# What Next?

Both AI Chatbots and AI Agents are very powerful and with frameworks such as Langchain, AI-based applications can be easily built to help businesses move to the next stages of their growth.

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